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# Survey-Based Case Study of Supply Chain Management (SCM) in Construction Industries

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***Abstract***: *The construction industry is another sector that substantially contributes to economic growth and is also a very fragmented industry that is vulnerable to material management inefficiencies. Supply Chain Management (SCM) provides the systematic approaches to eliminate delays, wastage and costs, but it is not widely applied in the construction companies of regions in India. This paper gives a survey-based case study research of construction firms in Maharashtra with special consideration given to Bindra Steel Pvt. Ltd. The research uses various inventory control methods, including ABC, HML, FSN, VED, SDE analysis, and codification system to measure procurement, storage, consumption and criticality of construction materials. Findings indicate that material management constitutes about 75 percent of overall cost of production and as such, it is the most substantial area that could be considered in reduction of costs. It is also revealed that scientific codification enhances traceability and decreases retrieval delays, and inventory classifications point out the important items that need to be carefully monitored. The research concludes that with structured SCM practices, cost efficiency, accuracy of forecasting and competitiveness of construction firms can be improved. The suggested recommendations are the implementation of digital inventory solutions, education, and active purchases programs, specific to the regional construction industry.*

***Keywords:*** *Supply Chain Management, Construction Industry, Material Management, ABC Analysis, Maharashtra, Inventory Control.*

## 1. INTRODUCTION

The construction sector is among the most significant in terms of national development because it is very instrumental in infrastructure development, generation of jobs, and economic development. It has a wide range of activities dealing with residential, commercial, industrial and infrastructure projects [1]. This industry is also one of the largest employers in India which attracts contractors, suppliers, architects, engineers and clients. Nevertheless, even such an important and large industry as the construction one is characterized by fragmentation and project-focused business, short-term relations, and non-standardization [2]. These structural challenges often give rise to inefficiencies like projects taking longer time to be completed, cost increase, and reduction in quality that consequently impact the productivity and competitiveness on the whole [3-6].

Supply Chain Management (SCM) is a concept that has gained relevancy in order to deal with these inefficiencies. The term SCM in the construction industry can be defined as a method of the systematic planning, coordination, and control of procurement, material flow, logistics, and information in the project lifecycle [7]. When properly implemented, SCM can lead to a better coordination of many stakeholders, minimization of wastage, improved inventory management, and better project delivery schedules. The SCM also offers a chance at the efficiency improvement, accountability and alignment with the client expectations by converting the traditional siloed operations into a single framework structure [8].

Irrespective of these benefits, very few SCM practices have been adopted in the Indian construction industry. Most local companies are still using traditional approaches to procurement and manual management of materials, and the utilization of digital technologies and scientific inventory management is not popular. This has led to problems of lack of clear communication between contractors and suppliers, inability to predict project schedule, excessive inventory prices, and lack of systematic application of methods like ABC, VED, or FSN analysis. As such, there is no efficiency of material management despite the fact that it is a significant part of the construction costs [9].

Maharashtra is an Indian state that is one of the most industrialized and urbanized and has seen a booming construction and real estate development in the state in the form of districts such as Pune, Nashik, Aurangabad, Jalna and Nagpur. However, most companies in these areas still do their work in the same manners and lack integration and scientific management. Such challenges as low use of technology, shortage of qualified supply chain experts, and untrustworthy networks of suppliers present significant challenges to efficient management of materials [10].

It is against this backdrop that this study explores the SCM application in the construction industry by conducting a survey of the local companies, and a detailed case study on Bindra Steel Pvt. Ltd. an intermediate to large construction firm in Maharashtra. Through implementation of inventory management methods including ABC, HML, VED, FSN and SDE analysis and material codification, the research aims to determine the effects of systematic SCM activities on procurement performance, cost management and project performance. The results show that material management contributes almost three-quarters of the overall cost of the project making it the most important area where the cost can be cut. The study notes that structured SCM practice would be useful in achieving high accuracy of forecast, ease of coordination and timely project completion, hence increasing efficiency and competitiveness in construction industry [11].

## 2. RESEARCH METHODOLOGY

The current research has used descriptive and analytical research design that combines survey and case study research in order to explore the practices involved in supply chain management in the construction sector. Two major sources of information were used to gather data. Structured questionnaires and interviews with the procurement managers, engineers, and store officers in regional construction firms were used to get primary data. This was complemented by the secondary data which was based on company records, inventory logs and company policy manuals which contained more information on the procurement systems and material management practices [12].

In the case study element, Bindra Steel Pvt. Ltd., a medium to large sized construction firm in Maharashtra that deals in structural fabrication and construction that is project-based was chosen. The company was selected due to the engagement in massive material procurement and inventory management, which makes it reflective of the problems that other companies in the area are experiencing.

To determine the effectiveness of managing materials and procurements, a number of scientific methods were used such as the ABC level, HML level, FSN, VED level, SDE level, SOS level, material codification level and detailed cost breakdown level. These techniques offered a systematic way to categorize materials based on their contribution to costs, single unit price, frequency of consumption, the challenge of procurement as well as their importance and also made it possible to evaluate the advances made through the systematic codification. These three components of surveys, insights of case studies, and analytical tools provided a thorough insight into supply chain practices within the construction sector [13-19].

## 3. RESULTS AND ANALYSIS

Based on the survey conducted regarding supply chain management it was observed that many of the construction companies does not use specific tools or controls for material management.

### 3.1 Criteria for Supplier Selection

According to the survey responses, the most important factor taken into consideration by companies in choosing suppliers is quality (80.2%). This demonstrates that companies consider it highly essential that materials should be of the necessary standards to prevent re-work and delay of projects. The next equally important criteria is cost (56.9%) and the delivery time (56.9), which shows that the industry is sensitive to project budgets and timely delivery. Reliability (50.3%) is also a determinant and it indicates that there should be consistency in supply. Such other reasons as supplier reputation (37.7%), and access to credit (26.9%) can be rated as secondary, yet still, relevant, particularly in the case of smaller companies that have to cope with cash flow limitations. Such results suggest that affordability is a significant aspect, but the quality of the materials is the leading parameter, which construction companies use when evaluating suppliers.

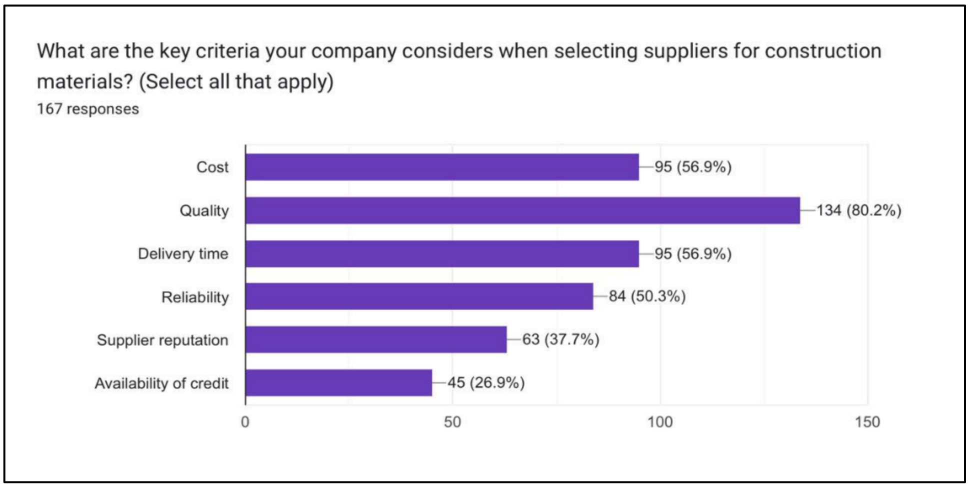


*Figure 1: Key Criteria Considered When Selecting Suppliers for Construction Materials*

### 3.2 Challenges in Procurement and Material Management

The survey indicates a number of issues encountered by companies in procurement and material management. Material shortages are considered the most frequently reported problem (61.7%), and they impact the project schedule greatly, causing uncertainty. Other significant barriers include supplier reliability (56.9%), and price fluctuations (52.1) that evidence that construction projects are prone to external disruption of their supply chain and market instability. Procurement is further complicated by quality problems (42.5%) whereby it requires regular checks and replacements. Another challenge is the lead times (32.3) which is normally due to transportation delays or inefficiency of the vendor. Even though fewer numbers indicated technology related problems, the results indicate that lack of digital solutions has an indirect effect on inefficiencies. All these problems prove that in case of the lack of organized supply chain systems, companies have a constant risk of obtaining and using materials.

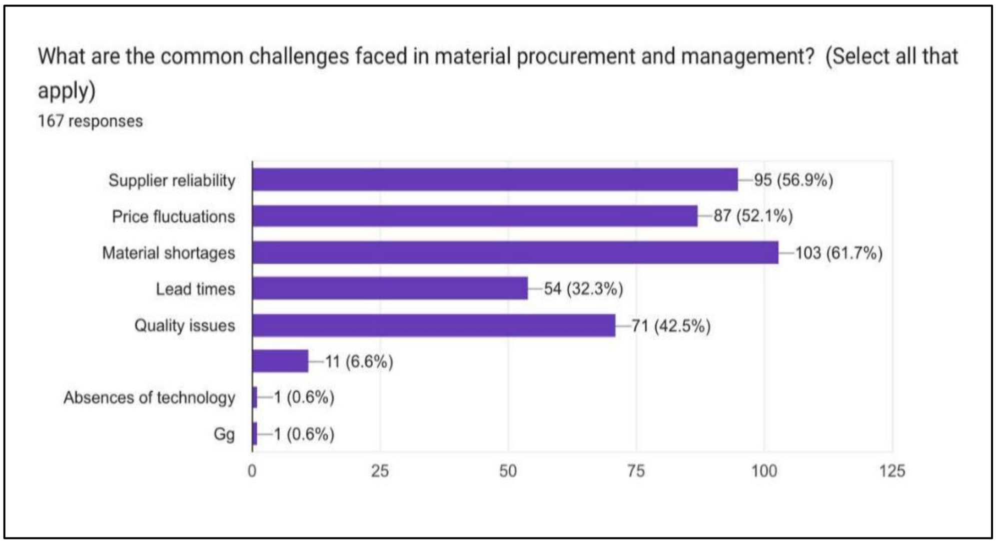


*Figure 2: Common Challenges Faced in Material Procurement and Management*

### 3.3 Challenges in Just-in-Time (JIT) Implementation

The research also investigated the challenges that the companies encounter when using Just-in-Time (JIT) inventory systems. The most frequent challenge is supplier delays (62.9%), which compromises the efficiency of JIT due to the disruption of supply flow and the establishment of bottlenecks at the site level. The unpredictable demand (45.5%) and storage problems on site (45.5%) are also major challenges. These points to the challenge in matching the supply and variable construction demand very accurately and the scarcity of storage facilities at most locations. Moreover, coordination between various suppliers (36.5%) appeared to be another issue because JIT demands delivery schedules coordination among the vendors. An extremely low proportion (3.6%) mentioned other operation challenges. All in all, the findings illustrate that although JIT has the ability to lower the cost of holding inventory, application of JIT to the construction industry is limited to high degree due to supplier reliability and uncertainty in demand.

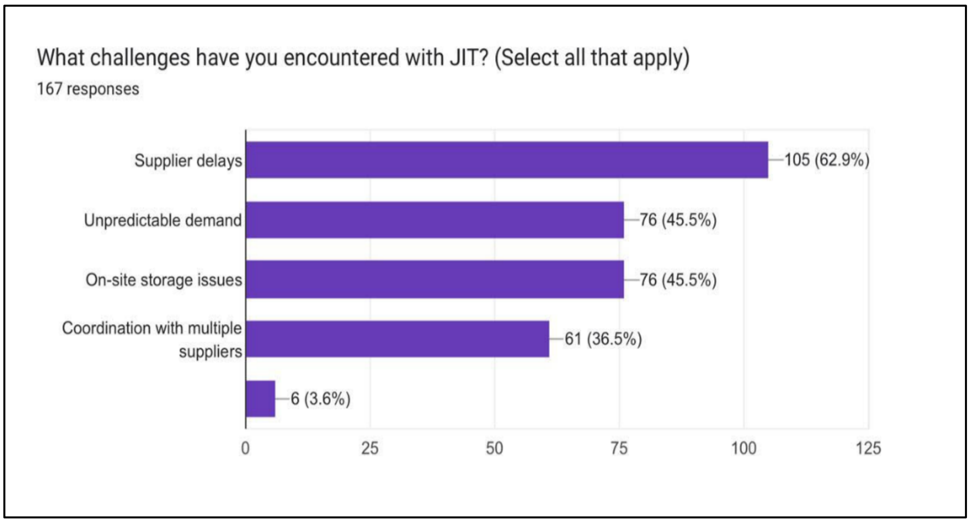


**Figure 3: Challenges Encountered with JIT in Construction Projects**

### 3.4 Practices of Material Tracking and Waste Management.

The questionnaire shows that a high dependency is on the old fashioned ways of monitoring material usage. The use of manual tracking (32.9) and spreadsheet software such as Excel (24.6) is combined (57.5%). This means that there is a predominant culture of reactive, paper based, or simple digital record-keeping, which is susceptible to human error and reporting delays. Only forty-one point three make use of specialized software (e.g. ERP systems) and scan tech, so it is not yet standard in the industry to have a sophisticated and integrated digital solution to real-time inventory management. This is an unavailability of technology and this is probably one of the reasons behind the inefficient inventory control, and wastage.

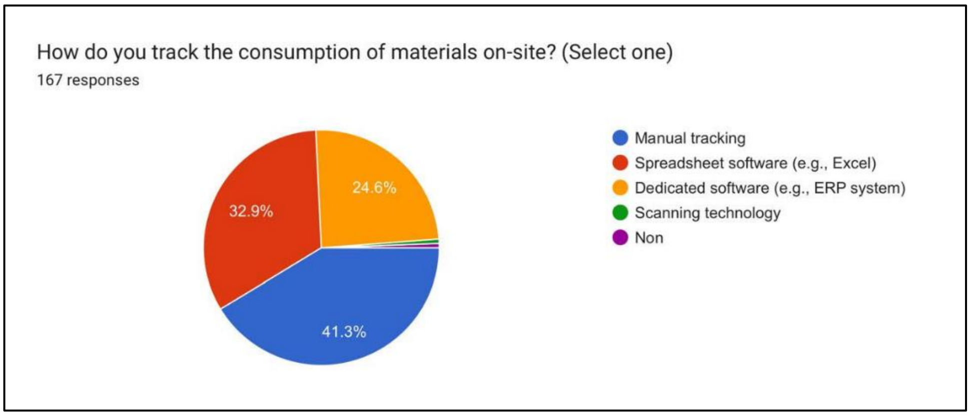


***Figure 4: Tracking Consumption Methods On-Site.***

Over-ordering is the main cause of material wastes as noted by 34.1% percent of the respondents. This is the direct result of bad planning and forecasting of the inventory which is usually made worse by the manual tracking methods mentioned earlier. Other important causes are destruction during handling (18%), ineffective utilization of the material (17.4%), which indicates the problem in the material handling process on the site and human staff management. Ineffective materials (16.6), incorrect storage (11.4) also bring waste on board, which supports the essential significance of supplier quality and on-site logistics.

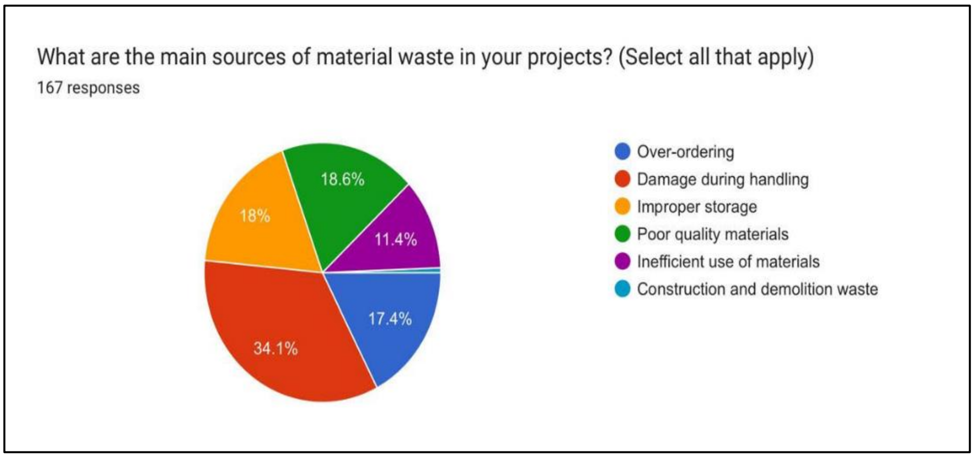


***Figure 5: Primary Material Waste in Projects.***

The information regarding the amount of waste is alarming. Although 44.9% of businesses claim to wasted below 5 percent of the materials, a large proportion (55.1) of them report wastes of 5 percent and above. It is worth noting that 16.2% of firms use between 10-20% of the materials, and 4.9% use over 20%. These statistics constitute a significant loss of project funds and point to a significant dimension of the possible value of cost-saving and operational optimization in terms of enhanced planning and control mechanisms.

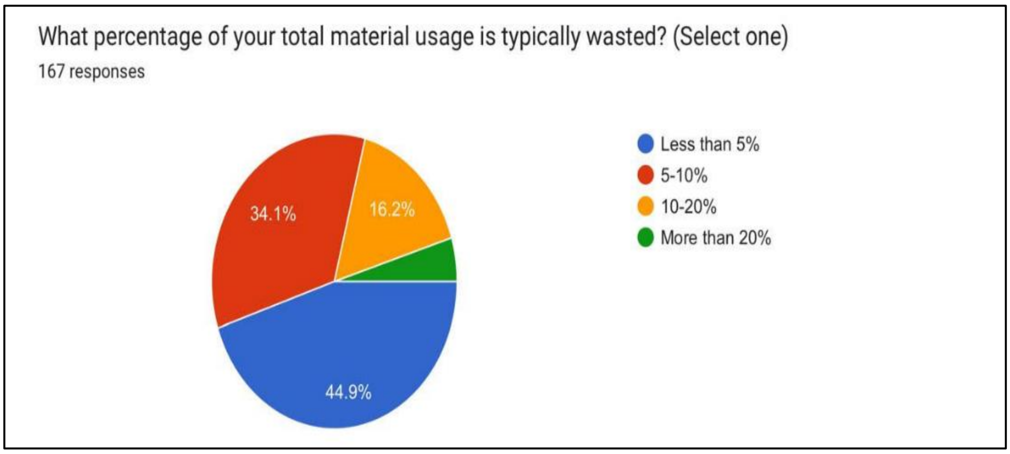


***Figure 6: Averaged Percentage of total material wasted.***

**3.5 Inventory Analysis Techniques Adoption.**

The findings of the survey demonstrate a very obvious trend of using formal inventory analysis methods. Regarding each and every approach interrogated, namely JIT, ABC, FSN, VED, SOS, as well as GOLF, the answer is remarkably close:

1. Largepercentages (approximately 65-75) of firms are not applying these methods.
2. Only a minor minority (approximately 15-20 percent) is using them.
3. The other minor segment (~8-10) is looking at implementation.

This consistency of trends shows that the absence of organized analytical methods of inventory management in the surveyed construction industry is widespread. The high Noes indicate that the majority of firms use experience instead of data-driven classification when making such critical decisions as inventory management and procurement planning and the choice of suppliers.

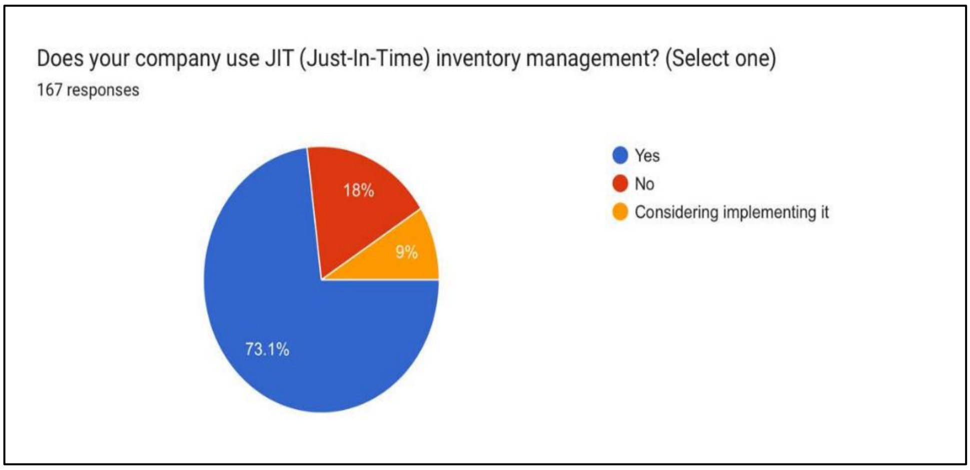


***Figure 7: JIT, ABC, FSN, VED, SOS and GOLF Analyses Adoption.***

The use of ABC Analysis is promising to the minority of firms who use it. The high percentage includes both A (High-value, low-quantity) items (53.3%), and B items (Moderate-value, moderate-quantity) items (58.7%). This enables them to exercise strict measures on the items that take the highest proportion in the budget. Nonetheless, 28.1 percent of them classify C (Low-value, high-quantity) items, which implies that even among adopters the system might not be implemented in all aspects and some efficiencies might not be taken off the table.

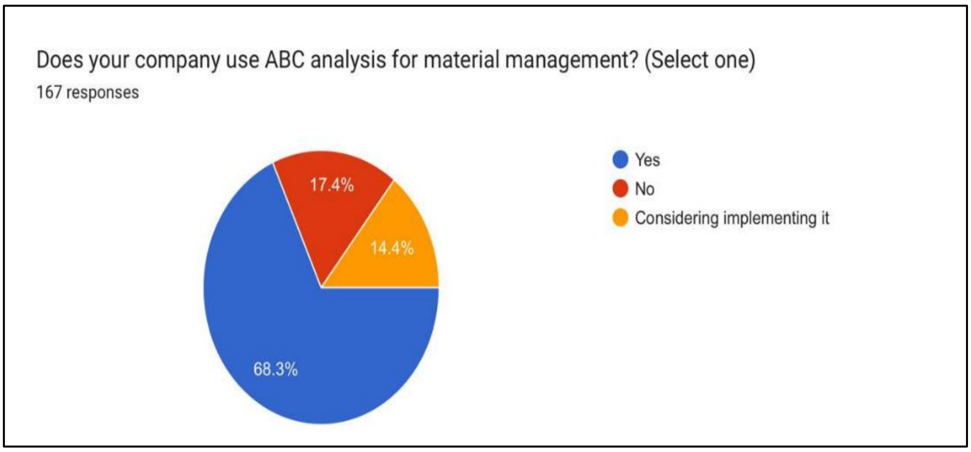


***Figure 8: How to use the Analysis of ABC (Among Users)***

Among the small number of users of FSN analysis, it has a practical impact: 43.1% of them use it when they have to review fast-moving products frequently, and 42.5% use it when they need to dispose of or redistribute non-moving products. This assists in maximising inventory turnover, as well as capital and space release.

In a similar way, in case of SOS analysis, users majorly use it to order bulk of the seasonal materials (47.9%), to continually order the items of steady demand (34.7%), which proves its usefulness in matching the procurement strategy with demand trends.

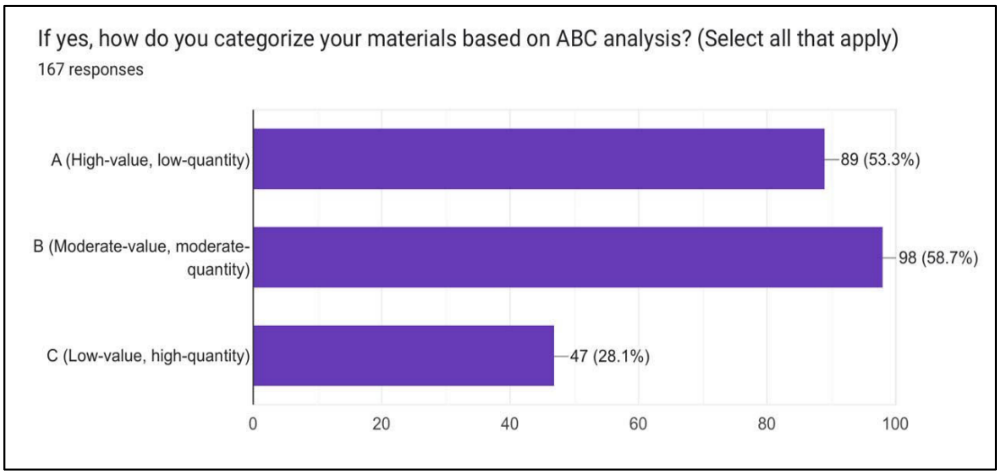


***Figure 9: The effects of FSN and SOS Analyses (Among the users).***

## 4. CONCLUSION

This paper was aimed at exploring the as follows to understand supply chain management practices in the construction industry; a survey analysis and a case study of Bindra Steel Pvt. Ltd. The results clearly show that the material management is the most important element of construction operations taking almost three-fourths of the total project costs. Although it is an important factor, the survey showed that the use of the traditional and manual procurement, tracking, and waste management procedures by firms has not been eliminated, and few systematic methods of inventory analysis have been implemented. The choice of suppliers was demonstrated to be based on the quality of material and then on the cost and delivery time, which points to the relevance of reliability and affordability in the choice of the supplier. Nevertheless, companies still experience endemic issues like shortages of materials, delays with suppliers, fluctuation of prices and quality. The lack of reliability of suppliers and uncertainty in demand discourages the efforts to introduce the latest techniques such as Just-in-Time (JIT) usage in construction. Material wastes, which tend to be caused by over-ordering, ineffective planning, and handling inefficiency were observed to be at a very high rate with more than half of the firms recording above five percent wastages. Another shortcoming that was found in the study was a general lack of principles of formal inventory control that even practices like ABC, FSN, VED, and SOS were practiced by a very small proportion of firms. These methods were useful in the allocation of the important materials where they were used, decreasing the non-movers, and matching the demand cycles with the procurement approaches. The case study also revealed that the material management traceability level, accuracy and efficiency can be greatly enhanced with the help of codification of materials and structured classification systems.

**REFERENCEs**


1. Studer, W., & De Brito Mello, L. (2021). Core Elements Underlying Supply Chain Management in the Construction Industry: A Systematic Literature Review. *Buildings*. https://doi.org/10.3390/buildings11120569
2. Badi, S., & Murtagh, N. (2019). Green supply chain management in construction: A systematic literature review and future research agenda. *Journal of Cleaner Production*. https://doi.org/10.1016/J.JCLEPRO.2019.03.132
3. Olsson, F. (2000). Supply Chain Management in the Construction Industry.
4. Nguyen, D., & Le, P. (2022). Twenty-year application of logistics and supply chain management in the construction industry. *Construction Management and Economics*, 40, 796 - 834. https://doi.org/10.1080/01446193.2022.2110273
5. Kosanoglu, F., & Kus, H. (2021). Sustainable supply chain management in construction industry: a Turkish case. *Clean Technologies and Environmental Policy*, 23, 2589 - 2613. https://doi.org/10.1007/s10098-021-02175-z
6. Liao, L., Yang, C., & Quan, L. (2022). Construction supply chain management: A systematic literature review and future development. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2022.135230
7. Kim, S., & Nguyen, V. (2020). Supply chain management in construction: critical study of barriers to implementation. *International Journal of Construction Management*, 22, 3148 - 3157. https://doi.org/10.1080/15623599.2020.1843768
8. Aloini, D., Dulmin, R., Mininno, V., & Ponticelli, S. (2012). Supply chain management: a review of implementation risks in the construction industry. *Bus. Process. Manag. J.*, 18, 735-761. https://doi.org/10.1108/14637151211270135
9. Papadopoulos, G., Zamer, N., Gayialis, S., & Tatsiopoulos, I. (2016). Supply Chain Improvement in Construction Industry. *Universal journal of management*, 4, 528-534. https://doi.org/10.13189/UJM.2016.041002
10. Wibowo, M., Handayani, N., & Mustikasari, A. (2018). Factors for implementing green supply chain management in the construction industry. *Journal of Industrial Engineering and Management*. https://doi.org/10.3926/jiem.2637
11. Han, Y., Yan, X., & Piroozfar, P. (2022). An overall review of research on prefabricated construction supply chain management. *Engineering, Construction and Architectural Management*. https://doi.org/10.1108/ecam-07-2021-0668
12. Fauzi, M., Hasim, S., & Mustafa, M. (2018). Supply Chain Management (SCM) on Industrialised Building System (IBS) in Construction Industry: Supplier Perspective., 445-454. https://doi.org/10.1007/978-981-13-0074-5_43
13. Studer, W., & De Brito Mello, L. (2021). Core Elements Underlying Supply Chain Management in the Construction Industry: A Systematic Literature Review. *Buildings*. https://doi.org/10.3390/buildings11120569
14. Nguyen, D., & Le, P. (2022). Twenty-year application of logistics and supply chain management in the construction industry. *Construction Management and Economics*, 40, 796 - 834. https://doi.org/10.1080/01446193.2022.2110273
15. Liao, L., Yang, C., & Quan, L. (2022). Construction supply chain management: A systematic literature review and future development. *Journal of Cleaner Production*. https://doi.org/10.1016/j.jclepro.2022.135230
16. Kanda, A. (2014). Supply chain management in construction. https://doi.org/10.1680/scmic.42810
17. Cigolini, R., Gosling, J., Iyer, A., & Senicheva, O. (2020). Supply chain management in construction and engineer-to-order industries. *Production Planning & Control*, 33, 803 - 810. https://doi.org/10.1080/09537287.2020.1837981

18. Papadopoulos, G., Zamer, N., Gayialis, S., & Tatsiopoulos, I. (2016). Supply Chain Improvement in Construction Industry. *Universal journal of management*, 4, 528-534. https://doi.org/10.13189/UJM.2016.041002
19. Han, Y., Yan, X., & Piroozfar, P. (2022). An overall review of research on prefabricated construction supply chain management. *Engineering, Construction and Architectural Management*. https://doi.org/10.1108/ecam-07-2021-0668